DATA STORY Alexander Zhuk Date: 11/30/2025

Deployed Shiny App URL

https://019ab2f0-c599-c119-a1cb-0ce4d5acc432.share.connect.posit.cloud/

Code Repository URL

https://github.com/AZhuk30/ShinyApp

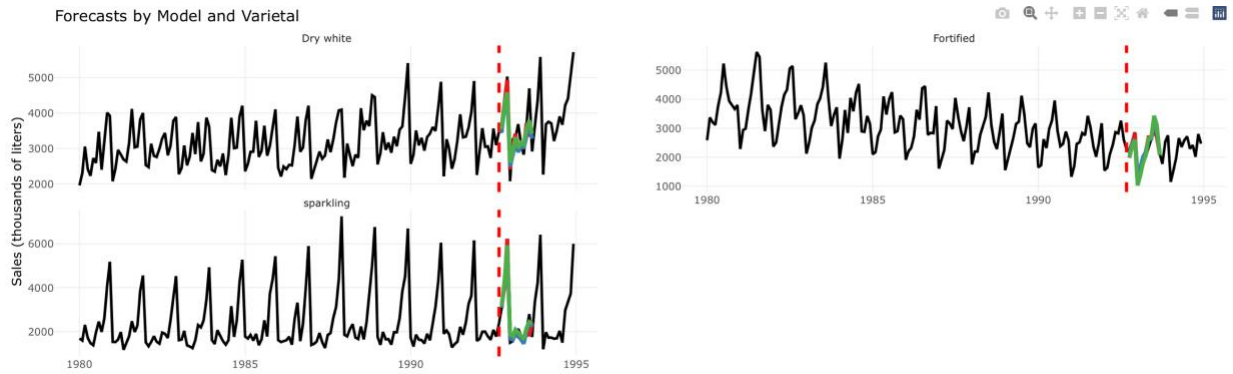
Varietal-Specific Forecasting: Why One Model Doesn't Fit All Australian Wines

The Finding

Looking at Australian wine sales from 1980-1994, I found something that surprised me: not all wines behave the same way seasonally, so they need different forecasting approaches. Sparkling and Dry white wines have massive December spikes and Sparkling can jump to over 6,000 liters when normal months are under 4000 liters. These extreme swings need ARIMA models that can handle seasonal differencing. But Fortified wine stays pretty stable year-round, so a simpler ETS model works just as well as ARIMA. For wineries trying to avoid stockouts in December or sitting on too much inventory in February, matching the model to each wine's behavior actually matters.

What the Data Shows

Figure 1: Comparative Forecasts Across Models and Varietals



Reproduction settings: App URL: https://019ab2f0-c599-c119-a1cb-0ce4d5acc432.share.connect.posit.cloud/ | Varietals: Fortified, Sparkling, Dry white | Models: TSLM, ETS, ARIMA | Training Data: 80% | Forecast Horizon: 12 months

The patterns are pretty different across wine types. Sparkling wine goes significantly higher in December as sales hit over 6,000 liters, which is more than triple a normal month. Dry white has strong seasonal peaks too, consistently around 5,000 liters. But Fortified wine? It's way more stable, staying between 2,000-4,500 liters without those massive swings. The green shaded areas after the red line show the forecast prediction intervals—you can see how each model handles uncertainty differently.

Figure 2: Validation Accuracy Metrics

Accuracy Metrics

Fortified	ETS	Validation	259.51	209.12	9.36
Fortified	ARIMA	Validation	277.74	211.04	9.1
sparkling	ARIMA	Validation	317.42	229.39	10.46
sparkling	ETS	Validation	342.41	253.83	11.76
Fortified	TSLM	Validation	349.79	282.17	12.39
sparkling	TSLM	Validation	369.81	284.59	14.04
Dry white	ARIMA	Validation	496.76	378.35	9.74
Dry white	TSLM	Validation	525.73	419.26	10.82
Dry white	ETS	Validation	617.15	499.45	12.49

Reproduction settings: Same as Figure 1 | View: Model Results tab → Accuracy Metrics table, sorted by RMSE, Validation Search in Focus

- **Fortified wine**: ETS gets 259.51 RMSE, ARIMA gets 277.74. That's only about 7% difference. The ETS model (M,A,M form) works fine for Fortified's steadier patterns.
- **Sparkling wine**: ARIMA wins at 317.42 versus ETS at 342.41. That's about 8% better, but more importantly, ARIMA's seasonal differencing actually captures those December spikes that would otherwise wreck inventory planning.
- **Dry white**: ARIMA clearly performs better at 496.76 RMSE. Its drift component and seasonal structure handle Dry white's evolving patterns over the 15 years.

For Sparkling wine, the difference between ARIMA and ETS is about 25 liters per month in average error. When you're dealing with December demand over 6,000 liters, that percentage improvement could mean having stock when customers want it versus running out. For Fortified wine, there's basically no reason to use the complicated ARIMA model. ETS works just as well and runs faster. You don't need fancy seasonal differencing when the patterns are already stable. I'd say split forecasting by how volatile each wine type is. Use ARIMA for the high-volatility stuff like Sparkling and Dry white where those seasonal patterns are intense. Stick with ETS for stable products like Fortified where the extra complexity doesn't buy you anything.

The model specs I ended up with were:

- **Fortified**: ETS(M,A,M) - multiplicative error, additive trend, multiplicative seasonality
- **Sparkling**: ARIMA(0,0,1)(0,1,1)[12] - seasonal differencing with 12-month period
- **Dry white**: ARIMA(1,0,0)(0,1,1)[12] w/ drift - autoregressive component plus seasonal structure